

1 Osnove teorije

1.1 BCS teorija pri končni temperaturi

BCS model, kjer je $\xi_{\mathbf{k},\sigma} = \epsilon_{\mathbf{k},\sigma} - E_F$ in $k := (\mathbf{k}, \uparrow)$, $-k := (-\mathbf{k}, \downarrow)$:

$$H = \sum_k \xi_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) - V \sum_{k,k'} c_k^\dagger c_{-k}^\dagger c_{-k'} c_{k'} \quad (1)$$

Naredimo približek povprečnega polja, definiramo $b_k = c_k c_{-k}$ in $b_k = \langle b_k \rangle + \delta b_k = F_k + \delta b_k$, sledi (do $\mathcal{O}(\delta b_k^2)$)

$$b_k^\dagger b_{k'} \simeq F_k^* F_{k'} + F_k^* \delta b_{k'} + F_{k'} \delta b_k^\dagger = F_k^* b_{k'} + F_{k'} b_k^\dagger - F_k^* F_{k'} = \langle b_k^\dagger \rangle b_{k'} + b_k^\dagger \langle b_{k'} \rangle - \langle b_k^\dagger \rangle \langle b_{k'} \rangle \quad (2)$$

Iz česar sledi:

$$H = \sum_k (c_k^\dagger \quad c_{-k}) \begin{pmatrix} \xi_k & -\Delta_k \\ -\Delta_k^* & -\xi_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} + \frac{|\Delta|^2}{V} + \sum_k \xi_k \quad (3)$$

Ob definiciji $\Delta_k = \sum_{k'} V(k - k') \langle c_{k'} c_{-k'} \rangle$, vzamemo še lokalno interakcijo, zato je $V(q) \simeq V$ in $\Delta_k \simeq \Delta$. Naredimo Bogoljubov rotacijo, privzamemo $u_k, v_k \in \mathbb{R}$ in simetrijo okoli E_F : $u_k = u_{-k}$ ter $v_k = v_{-k}$. Iz CCR sledi $u_k^2 + v_k^2 = 1$. Kvazidelci so definirani z rotacijo

$$\begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix} \quad (4)$$

nov Hamiltonjan pa postane

$$H = \sum_k (\alpha_k^\dagger \quad \alpha_{-k}) \begin{pmatrix} \xi_k u_k^2 + u_k v_k (\Delta_k + \Delta_k^*) - \xi_k v_k^2 & -2\xi_k u_k v_k + \Delta_k u_k^2 - \Delta_k^* v_k^2 \\ -2\xi_k u_k v_k - \Delta_k v_k^2 + \Delta_k^* u_k^2 & \xi_k v_k^2 - \xi_k u_k^2 + u_k v_k (\Delta_k + \Delta_k^*) \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix}, \quad (5)$$

kjer zahtevamo, da so izvendiagonalni členi enaki 0. Iz CCR zahteve lahko vzamemo parameterizacijo (tu sem vzel kompleksno nastavimo, da je $\Delta = \Delta^* e^{i2\phi_k}$):

$$v_k = \sin\left(\frac{\theta_k}{2}\right) e^{i\phi_k} \quad (6)$$

$$u_k = \cos\left(\frac{\theta_k}{2}\right) \quad (7)$$

$$u_k v_k = \cos\left(\frac{\theta_k}{2}\right) \sin\left(\frac{\theta_k}{2}\right) = \frac{1}{2} \sin \theta_k \quad (8)$$

$$u_k^2 - v_k^2 = \cos^2\left(\frac{\theta_k}{2}\right) - \sin^2\left(\frac{\theta_k}{2}\right) = \cos \theta_k \quad (9)$$

skupaj dobimo (od tu naprej $\Delta := |\Delta|$):

$$\xi_k \sin(\theta_k) = \Delta \cos(\theta_k) \quad (10)$$

kar fiksira

$$\tan(\theta_k) = \frac{|\Delta_k|}{\xi_k}. \quad (11)$$

Tako iz trikotnika preberemo

$$\sin \theta_k = \frac{|\Delta_k|}{E_k} = \frac{|\Delta_k|}{\sqrt{\xi_k^2 + |\Delta_k|^2}} \quad (12)$$

$$\cos \theta_k = \frac{\xi_k}{E_k} = \frac{\xi_k}{\sqrt{\xi_k^2 + |\Delta_k|^2}} \quad (13)$$

$$(14)$$

kjer je $E_k = \pm \sqrt{\xi_k^2 + |\Delta_k|^2}$ energija kvazidelcev:

$$H = \sum_k E_k \alpha_k^\dagger \alpha_k - E_k \alpha_{-k}^\dagger \alpha_{-k} + \xi_k + \frac{|\Delta|^2}{V}. \quad (15)$$

V termičnem stanju so diagonalni delci porazdeljeni po

$$\langle \alpha_k^\dagger \alpha_k \rangle = \frac{1}{1 + \exp(E_k/T)} = \frac{1}{2} - \frac{1}{2} \tanh\left(\frac{E_k}{2T}\right). \quad (16)$$

Uporabni sta zvezi $f(-E/T) = 1 - f(E/T)$ in $1 - 2f(E/T) = \tanh(E/(2T))$. V definiciji ureditvenega parametra je

$$\langle c_k c_{-k} \rangle = \left\langle \left(u_k \alpha_k - v_k \alpha_{-k}^\dagger \right) \left(u_k \alpha_{-k} + v_k \alpha_k^\dagger \right) \right\rangle = u_k v_k \left(1 - \langle \alpha_{-k}^\dagger \alpha_{-k} \rangle - \langle \alpha_k^\dagger \alpha_k \rangle \right) = \frac{1}{2} \sin \theta_k \tanh\left(\frac{\beta E_k}{2}\right) \quad (17)$$

iz česar sledi enačba za režo pri $T > 0$ (seštevamo po k okoli E_F):

$$\Delta = \frac{V}{2} \sum_k' \sin \theta_k \tanh\left(\frac{E_k}{2T}\right) = \frac{V\Delta}{2} \sum_k \frac{\tanh\left(\frac{1}{2T} \sqrt{\Delta^2 + \xi_k^2}\right)}{\sqrt{\Delta^2 + \xi_k^2}} \quad (18)$$

1.2 Minimalni model ekscitonskega izolatorja pri končni temperaturi

Pri izolatorju za minimalni model vzamemo dve orbitali (a, b) model tesne vezi v vsaki posebej in lokalno Coulombovo interakcijo na mestu j med elektronskima gostotama v posameznih pasovih

$$H = H_0 + H_V = \sum_{i,j} \left(\epsilon_a^{(0)} \delta_{ij} + t_{ij}^{(a)} \right) c_{a,i}^\dagger c_{a,j} + \sum_{i,j} \left(\epsilon_b^{(0)} \delta_{ij} + t_{ij}^{(b)} \right) c_{b,i}^\dagger c_{b,j} + V \sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \quad (19)$$

$$= \sum_k \varepsilon_{a,k} c_{a,k}^\dagger c_{a,k} + \sum_k \varepsilon_{b,k} c_{b,k}^\dagger c_{b,k} + V \sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \quad (20)$$

najprej diagonaliziramo vsako orbitalo posebej, kar da gole disperzije (vezavni energiji nastavimo na $\pm G/2$) (delamo v dveh dimenzijah in mrežna konstanta $a_0 = 1$):

$$\varepsilon_{a(b),k} = \pm \frac{G}{2} - 2t_{a(b)}(\cos(k_x) + \cos(k_y)), \quad (21)$$

v tem primeru je najmanjša razdalja med pasovoma $G_0 = G - 4(t_a - t_b)$ oziroma $G_0 = G - 8t_a$, če je $t_b = -t_a$. Za interakcijo spet naredimo približek povprečnega polja:

$$c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \approx \langle c_{a,j}^\dagger c_{a,j} \rangle c_{b,j}^\dagger c_{b,j} + \langle c_{b,j}^\dagger c_{b,j} \rangle c_{a,j}^\dagger c_{a,j} - \langle c_{a,j}^\dagger c_{b,j} \rangle c_{b,j}^\dagger c_{a,j} - \langle c_{b,j}^\dagger c_{a,j} \rangle c_{a,j}^\dagger c_{b,j} + \text{konst.} \quad (22)$$

od tod sledi

$$H_V^{\text{MF}} = V n_b \sum_j c_{a,j}^\dagger c_{a,j} + V n_a \sum_j c_{b,j}^\dagger c_{b,j} - V \phi \sum_j c_{b,j}^\dagger c_{a,j} - V \phi^* \sum_j c_{a,j}^\dagger c_{b,j} + NV |\phi|^2 \quad (23)$$

kjer smo definirali:

$$n_a = \langle c_{a,j}^\dagger c_{a,j} \rangle = \frac{1}{N} \sum_k \langle c_{a,k}^\dagger c_{a,k} \rangle \quad (24)$$

$$n_b = \langle c_{b,j}^\dagger c_{b,j} \rangle = \frac{1}{N} \sum_k \langle c_{b,k}^\dagger c_{b,k} \rangle \quad (25)$$

$$\phi = \langle c_{a,j}^\dagger c_{b,j} \rangle = \frac{1}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle \quad (26)$$

$$\Delta = -\frac{V}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle, \quad \Delta^* = -\frac{V}{N} \sum_k \langle c_{b,k}^\dagger c_{a,k} \rangle \quad (27)$$

$$\Delta_k = \frac{1}{N} \sum_{k'} V(k - k') \langle a_k^\dagger, b_{k'} \rangle, \quad V(q) = \int e^{-i\mathbf{q} \cdot \mathbf{r}} V(\mathbf{r}) \, d\mathbf{r} \quad (28)$$

Uporabimo FT še na celotnem Hamiltonjanu

$$c_{a,j}^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{-i\mathbf{k} \cdot \mathbf{r}_j} c_{a,k}^\dagger \quad (29)$$

$$H^{\text{MF}} = \sum_k \left([\varepsilon_{a,k} + V n_b] c_{a,k}^\dagger c_{a,k} + [\varepsilon_{b,k} + V n_a] c_{b,k}^\dagger c_{b,k} + \Delta^* c_{a,k}^\dagger c_{b,k} + \Delta c_{b,k}^\dagger c_{a,k} \right) + \left(\frac{|\Delta|^2}{V} N - V N n_a n_b \right) \quad (30)$$

oziroma:

$$H^{\text{MF}} = \sum_k \begin{pmatrix} c_{a,k}^\dagger & c_{b,k}^\dagger \end{pmatrix} \begin{pmatrix} \epsilon_{a,k}^H & \Delta^* \\ \Delta & \epsilon_{b,k}^H \end{pmatrix} \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} + \text{konst.}, \quad (31)$$

kjer definiramo Hartree energiji $\epsilon_{a(b),k}^H = \varepsilon_{a(b),k} + V n_{b(a)}$. Transformacija v diagonalno / Bogoljubov bazo je podobna kot pri BCS, le da ne meša kreatorjev in anihilatorjev / ohranja število delcev:

$$\begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} \quad (32)$$

od tod po enakem postopku kot prej dobimo energiji kvazidelcev

$$E_{\alpha(\beta)}(k) = \frac{\epsilon_{a,k}^H + \epsilon_{b,k}^H}{2} \pm \sqrt{\left(\frac{\epsilon_{a,k}^H - \epsilon_{b,k}^H}{2} \right)^2 + |\Delta|^2} := \eta_k \pm \sqrt{\xi_k^2 + \Delta^2} \quad (33)$$

Za bookkeeping:

$$\left(\frac{\epsilon_a^H - \epsilon_b^H}{2} \right) \sin \theta_k + \Delta \cos \theta_k = 0 \implies \tan \theta_k = \frac{-2\Delta}{\epsilon_a^H - \epsilon_b^H} = \frac{-\Delta}{\xi_k} \quad (34)$$

$$u_k v_k = -\frac{\sin \theta_k}{2}. \quad (35)$$

Za samousklajeno enačbo zdaj potrebujemo

$$\langle c_{a,k}^\dagger c_{b,k} \rangle = \langle (u_k \alpha_k^\dagger + v_k \beta_k^\dagger) (-v_k \alpha_k + u_k \beta_k) \rangle = u_k v_k \left(\langle \beta_k^\dagger \beta_k \rangle - \langle \alpha_k^\dagger \alpha_k \rangle \right) = -\frac{1}{2} \sin \theta_k (f(E_\beta) - f(E_\alpha)) \quad (36)$$

od tod

$$\Delta = -\frac{V}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle = \frac{V}{N} \sum_k \frac{\Delta}{2\sqrt{\xi_k^2 + |\Delta|^2}} (f(E_\beta) - f(E_\alpha)) \quad (37)$$

Opomba, lahko definiramo tudi $n = n_a + n_b = \text{konst.} = 1$ in $n_a - n_b =: m$ od tod sledi

$$H^{\text{MF}} = \sum_k \left(\left[\varepsilon_{a,k} + \frac{V}{2} n - \frac{V}{2} m \right] c_{a,k}^\dagger c_{a,k} + \left[\varepsilon_{b,k} + \frac{V}{2} n - \frac{V}{2} m \right] c_{b,k}^\dagger c_{b,k} + \Delta^* c_{a,k}^\dagger c_{b,k} + \Delta c_{b,k}^\dagger c_{a,k} \right) \quad (38)$$

od koder je v 2D Hartree gap enak:

$$G^H = G - 4(t_a - t_b) - Vm. \quad (39)$$

Note: v praksi je $m < 0$ V dveh dimenzijah se Hartree pasova srečata na

$$\mu_B = \frac{V}{2} n - \frac{t_a + t_b}{t_a - t_b} \frac{G - Vm}{2}. \quad (40)$$

Zamik je enak $V/2$, ko sta osnovna pasova simetrična in je zasedenost polovična. V primeru simetričnih pasov sta disperziji kvazidelcev

$$E_{\alpha(\beta)}(k) - V/2 = \pm \sqrt{\left(\frac{\epsilon_{a,k} - \epsilon_{b,k} - Vm}{2}\right)^2 + |\Delta|^2}. \quad (41)$$

Velja še

$$n_a = \frac{1}{N} \sum_k (u_k^2 f(E_\alpha) + v_k^2 f(E_\beta)) \quad (42)$$

$$n_b = \frac{1}{N} \sum_k (v_k^2 f(E_\alpha) + u_k^2 f(E_\beta)) \quad (43)$$

$$u_k^2 = \frac{1}{2} \left(1 + \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}} \right) \quad (44)$$

$$v_k^2 = \frac{1}{2} \left(1 - \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}} \right) \quad (45)$$

1.3 Numerika/psevdokoda

1.4 Odprti sistemi in modeliranje kopeli

Born-Markov master enačbe in Redfield: ločimo kopel in sistem,

$$\rho_S(t) = \text{Tr}_B(\rho(t)) \quad (46)$$

Hamiltonjan razcepimo na $H = H_0 + H_I$, kjer je $H_0 = H_S \otimes \mathbb{I}_B + \mathbb{I}_S \otimes H_B$ razdeljen na podсистema, za interakcijski del pa privzamemo obliko $H_I = \sum_\alpha A_\alpha \otimes B_\alpha$. V interakcijski sliki je operator za evolucijo

$$U_0(t, t_0) = e^{-iH_0(t-t_0)} \quad (47)$$

$$\rho_I(t) = U_0^\dagger(t, 0) \rho(t) U_0(t, 0) = e^{iH_0 t} \rho(t) e^{-iH_0 t} \quad (48)$$

$$H_I(t) = U_0^\dagger(t, 0) H_I U_0(t, 0) = e^{iH_0 t} H_I e^{-iH_0 t} \quad (49)$$

Von Neumannova enačba je

$$\frac{d}{dt} \rho_I(t) = -i[H_I(t), \rho_I(t)] \quad (50)$$

$$\rho_I(t) = \rho_I(0) - i \int_0^t dt' [H_I(t'), \rho_I(t')] \quad (51)$$

vstavimo nazaj v komutator in se znebimo prostosnih stopenj kopeli:

$$\frac{d}{dt} \rho_S(t) = - \int_0^t ds \text{Tr}_B ([H_I(t), [H_I(t-s), \rho_S(t) \otimes \rho_B]]) \quad (52)$$

V interakcijskem členu so operatorji A_α napisani kot lastni operatorji H_S z energijami prehodov $\omega = E_b - E_a$

$$A_\alpha(\omega) = \sum_{E_b - E_a = \omega} |a\rangle \langle a| A_\alpha |b\rangle \langle b| \quad (53)$$

To zadošča relacijam $[H_S, A_\alpha(\omega)] = -\omega A_\alpha(\omega)$, $[H_S, A_\alpha^\dagger(\omega)] = \omega A_\alpha^\dagger(\omega)$ in $e^{iH_S t} A_\alpha(\omega) e^{-iH_S t} = e^{-i\omega t} A_\alpha(\omega)$, $e^{iH_S t} A_\alpha^\dagger(\omega) e^{-iH_S t} = e^{i\omega t} A_\alpha^\dagger(\omega)$. V interakcijski sliki potem velja

$$H_I(t) = \sum_{\alpha, \omega} e^{-i\omega t} A_\alpha(\omega) \otimes B_\alpha(t) = e^{iH_B t} B_\alpha e^{-iH_B t} \quad (54)$$

evolucija sledi kot

$$\dot{\rho}_S = \sum_{\omega, \omega'} \sum_{\alpha, \beta} e^{i(\omega' - \omega)t} \Gamma_{\alpha\beta}(\omega) \left(A_\alpha(\omega) \rho_S(t) A_\beta^\dagger(\omega') - A_\beta^\dagger(\omega') A_\alpha(\omega) \rho_S(t) \right) + \text{h.c.} \quad (55)$$

kjer so prehodne matrike med stanji

$$\Gamma_{\alpha\beta}(\omega) = \int_0^\infty ds e^{i\omega s} \text{Tr}_B (B_\alpha^\dagger(t) B_\beta(t-s) \rho_B) = \frac{1}{2} \gamma_{\alpha\beta}(\omega) + iS_{\alpha\beta}(\omega) \quad (56)$$

Evolucija pa ima strukturo

$$\frac{d\rho_S}{dt} = -i[H_{LS}, \rho_S(t)] + \mathcal{D}(\rho_S(t)) \quad (57)$$

z disipatorjem

$$\mathcal{D}(\rho_S) = \sum_{\omega} \sum_{\alpha, \beta} \gamma_{\alpha\beta}(\omega) \left(A_\beta(\omega) \rho_S A_\alpha^\dagger(\omega) - \frac{1}{2} \{A_\alpha^\dagger(\omega) A_\beta(\omega), \rho_S\} \right) \quad (58)$$

oz. v splošnem

$$\mathcal{D}(\rho_S) = \sum_k \gamma_k \left(L_k \rho_S L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho_S\} \right) \quad (59)$$

V stacionarnem stanju velja

$$\mathcal{L}\rho_{\text{ness}} = 0. \quad (60)$$

Notes za vektorizacijo in superoperatorje:

$$\mathcal{L}u_\mu^R = \lambda_\mu u_\mu^R \quad (61)$$

$$\text{vec}(ABC) = (A \otimes C^T) \text{vec}(B) \quad (62)$$

$$\text{Tr}(A^\dagger B) = \text{vec}(A)^\dagger \text{vec}(B) \quad (63)$$

$$\frac{d}{dt}|\rho\rangle = \left(-iH \otimes \mathbb{I} + i\mathbb{I} \otimes H^T + \sum_k L_k \otimes L_k^* - \frac{1}{2} \sum_k (L_k^\dagger L_k \otimes \mathbb{I} + \mathbb{I} \otimes L_k^T L_k^*) \right) |\rho\rangle \quad (64)$$

$$\langle \hat{O} \rangle(t) = \langle \mathbb{I} | \hat{O} \otimes \mathbb{I} | \rho(t) \rangle \quad (65)$$

Šibke sklopitve in perturbacijski pristopi

$$\mathcal{L} = -i[H, \cdot] + \epsilon \mathcal{D}[\cdot], \quad \rho = \rho_0 + \epsilon \rho_1 + \epsilon^2 \rho_2 + \dots \quad (66)$$

napišemo gostotno matriko sistema v lastni bazi, siagonalno, s klasičnimi populacijami

$$\rho_0 = \sum_n p_n |n\rangle \langle n|, \quad H|n\rangle = E_n |n\rangle \quad (67)$$

kjer velja v ničtem redu (by construction)

$$\mathcal{L}\rho_0 = -i[H, \rho_0] = 0 \quad (68)$$

V prvem redu so enačbe za NESS

$$-i[H, \rho_1] + \mathcal{D}(\rho_0) = 0 \quad (69)$$

kjer je $\mathcal{D}(\rho_0) = \sum_{\mu} (L_{\mu}\rho_0 L_{\mu}^{\dagger} - \frac{1}{2}\{L_{\mu}^{\dagger}L_{\mu}, \rho_0\})$ in od tod enačba za prehode med populacijami

$$\langle n | \mathcal{D}(\rho_0) | n \rangle = \sum_{\mu} \sum_m p_m |\langle n | L_{\mu} | m \rangle|^2 - \sum_{\mu} \sum_m p_n |\langle m | L_{\mu} | n \rangle|^2 \quad (70)$$

kot klasična populacijska dinamika z $R_{nm} = \sum_{\mu} |\langle n | L_{\mu} | m \rangle|^2$

$$\dot{p}_n = \epsilon \sum_m (R_{nm} p_m - R_{mn} p_n) \quad (71)$$

Izven diagonali elementi se razvijejo hitro na $\mathcal{O}(1)$, medtem ko se populacije razvijajo počasi na GGE manifoldu z $\mathcal{O}(\epsilon^{-1})$. Za ohranitev normalizacije mora veljati $\sum_n \dot{p}_n = 0$.

Integrabilni sistemi imajo makroskopsko število ohranjenih nabojev $[H_0, C_i] = 0$, pod unitarno časovno evolucijo se razvijejo v netermično stanje

$$\rho_{GGE} = \frac{1}{Z} e^{-\sum_i \lambda_i C_i} \quad (72)$$

$$\langle \psi_0 | C_i | \psi_0 \rangle = \text{Tr}(C_i \rho_{GGE}) \quad (73)$$

fiksira λ_i . UNitarna dinamika pripcira na GGE manifold, šibka sklopitev premika po GGE manifoldu:

$$\rho_0(t) = \rho_{\lambda(t)} = \frac{1}{Z(t)} \exp \left(- \sum_i \lambda_i(t) C_i \right) \quad (74)$$

EOM za ohranjene naboje je

$$\langle \dot{C}_i \rangle_{\lambda(\infty)} = \text{Tr}[C_i \mathcal{D}\rho] \simeq \epsilon \text{Tr}[C_i \mathcal{D}\rho_0] = 0 \quad \forall \lambda_i \quad (75)$$

Definiramo projektor na GGE manifold:

$$PX = - \sum_{ij} \frac{\partial \rho}{\partial \lambda_i} (\chi^{-1})_{ij} \text{Tr}[C_j X] \quad (76)$$

ki da dinamiko vzdolž manifolda:

$$\frac{\partial \rho_{\lambda}}{\partial t} = \sum_i \frac{\partial \rho_{\lambda}}{\partial \lambda_i} \dot{\lambda}_i = \epsilon P \mathcal{D}(\rho_{\lambda}) = -\epsilon \sum_{ij} \frac{\partial \rho_{\lambda}}{\partial \lambda_i} (\chi^{-1})_{ij} \text{Tr}[C_j \mathcal{D}(\rho_{\lambda})] \quad (77)$$

od koder sledi evolucija za koeficiente/potenciale λ_i :

$$\dot{\lambda}_i = -\epsilon \sum_j (\chi^{-1})_{ij} \text{Tr}[C_j \mathcal{D}\rho_{\lambda(t)}] \quad (78)$$

s susceptibilnostmi $\chi_{ij} = \langle C_i C_j \rangle - \langle C_i \rangle \langle C_j \rangle = -\text{Tr} \left[C_i \frac{\partial \rho}{\partial \lambda_j} \right]$.

2 Ekscitonski izolatorji in sklopitev s kopelmi

$$\text{Tr} \left[f \frac{1}{2} \{\rho, A\} \right] = \text{Tr} \left[\frac{1}{2} f \rho A + \frac{1}{2} f A \rho \right] = \frac{1}{2} \text{Tr} [A f \rho + f A \rho] \quad (79)$$

$$= \frac{1}{2} \text{Tr} [(A f + f A) \rho] = \frac{1}{2} \langle A f + f A \rangle \quad (80)$$

2.1 Osnovni modeli kopeli

2.2 Limita neskončne temperature

2.3 Fononske sklopitve

Najprej pogledamo, kako modificiran (Peierls tip) hopping integral in modificiran orbital energy vplivata na sklopitev. Model spremenimo: orbitali znotraj o.c. sta na enakem položaju in se deformirata vedno v isto smer, torej $\mathbf{u}_{a,i} = \mathbf{u}_{b,i} = \mathbf{u}_i$. Coulombski potencial se zaradi deformacije ne spremeni veliko. Hamiltonka za nedeformiran kristal ostane enaka, višje člene pa obravnavamo perturbativno:

$$H[\{\mathbf{u}\}] = \underbrace{\sum_{i,\gamma,\gamma'} \varepsilon_{\gamma\gamma',i}[\{\mathbf{u}\}] c_{\gamma,i}^\dagger c_{\gamma',i}}_{\text{on-site in CEF}} + \underbrace{\sum_{\langle i,j \rangle, \gamma} t_\gamma(|\mathbf{r}_{ij}|) c_{\gamma,i}^\dagger c_{\gamma,j}}_{\text{hopping, med orbitalami}} + \underbrace{V \sum_j n_{a,j} n_{b,j}}_{\text{Coulomb}} \quad (81)$$

Razdalje so:

$$\mathbf{r}_j = \mathbf{R}_j + \mathbf{u}_j, \quad (82)$$

oziroma razdalje med paroma (sosed):

$$\mathbf{r}_{ij} = \boldsymbol{\delta} + \mathbf{u}_j - \mathbf{u}_i \rightarrow |\mathbf{r}_{ij}| \simeq |\boldsymbol{\delta}| + \boldsymbol{\delta} \cdot (\mathbf{u}_j - \mathbf{u}_i) + \mathcal{O}(u^2) \quad (83)$$

Predpostavimo da na hopping vpliva samo sprememba razdalje:

$$t_\gamma(|\mathbf{r}_{ij}|) \simeq t_\gamma(|\boldsymbol{\delta}|) + \boldsymbol{\delta} \cdot (\mathbf{u}_j - \mathbf{u}_i) \left. \frac{dt_\gamma(x)}{dx} \right|_{x=|\boldsymbol{\delta}|=1} \quad (84)$$

Deformacija vpliva na vezavno energijo preko CEF, razvijemo preko deformacijskega tenzorja ($\varepsilon_{\mu\nu} = \frac{1}{2}(\partial_\mu u_\nu + \partial_\nu u_\mu)$ oziroma diskretno: $\varepsilon_{\mu\nu,j} = \frac{1}{4}(u_{j+\mu}^\nu - u_{j-\mu}^\nu + u_{j+\nu}^\mu - u_{j-\nu}^\mu)$ zgornji indeks je koordinata vektorja). Če je eno elektronska Hamiltonka na mestu j (el. - ion) (po μ, ν se seštevata po Einsteinu):

$$h_j = h_0 + \delta h_j + \mathcal{O}(\varepsilon^2), \quad \delta h_j = \frac{\partial U_{\text{el-ion}}}{\partial \varepsilon_{\mu\nu,j}} \varepsilon_{\mu\nu,j}, \quad (\mu, \nu) = (x, y) \quad (85)$$

in je v nedeformiranem stanju:

$$\varepsilon_\gamma^{(0)} = \langle \phi_\gamma | h_0 | \phi_\gamma \rangle \quad (86)$$

potem so popravki (lahko mešajo orbitali)

$$\delta \varepsilon_{\gamma\gamma'} = \left\langle \phi_\gamma \left| \frac{\partial U_{\text{el-ion}}}{\partial \varepsilon_{\mu\nu,j}} \right| \phi_{\gamma'} \right\rangle \varepsilon_{\mu\nu,j} = D_{\gamma\gamma'}^{\mu\nu} \varepsilon_{\mu\nu,j} \quad (87)$$

Sklopitvena Hamiltonka v realni bazi je tako ($\sum_{\langle ij \rangle} \rightarrow \sum_i \sum_{\delta=\{x,y\}}$)

$$H_{ep}^{(1)} = \sum_{i,\gamma,\gamma'} D_{\gamma\gamma'}^{\mu\nu} \varepsilon_{\mu\nu,i} c_{\gamma,i}^\dagger c_{\gamma',i} + \sum_{i,\delta,\gamma} t'_\gamma \boldsymbol{\delta} \cdot (\mathbf{u}_{i+\delta} - \mathbf{u}_i) \left(c_{\gamma,i}^\dagger c_{\gamma,i+\delta} + \text{h.c.} \right) \quad (88)$$

(note):

$$\sum_{\delta} \boldsymbol{\delta} \cdot (\mathbf{u}_{i+\delta} - \mathbf{u}_i) = \hat{x} \cdot (\mathbf{u}_{i+x} - \mathbf{u}_i) - \hat{x} \cdot (\mathbf{u}_{i-x} - \mathbf{u}_i) + \hat{y} \cdot (\mathbf{u}_{i+y} - \mathbf{u}_i) - \hat{y} \cdot (\mathbf{u}_{i-y} - \mathbf{u}_i) \approx 2(\varepsilon_{xx} + \varepsilon_{yy}) \quad (89)$$

Zdaj kvantiziramo odmike in prepišemo v k-prostor, specifično za 2D square lattice:

$$\mathbf{u}_j = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}, \lambda} \frac{1}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \boldsymbol{\epsilon}_{\mathbf{q}\lambda} e^{i\mathbf{q} \cdot \mathbf{R}_j} \left(b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger \right) \quad (90)$$

v prvem členu so sami členi oblike (od defomracijskega tenzorja, diskretna verzija)

$$u_{j+\delta}^\mu - u_{j-\delta}^\mu = \frac{2i}{\sqrt{N}} \sum_{\mathbf{q}, \lambda} \frac{\epsilon_{\mathbf{q}\lambda}^\mu}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \sin(q_\delta) e^{i\mathbf{q} \cdot \mathbf{R}_j} (b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger). \quad (91)$$

v drugem pa členi ($\delta = \mu = \{x, y\}$, glej note):

$$\frac{t'_\gamma}{\sqrt{N}} \sum_{\mathbf{q}, \mathbf{k}', \mathbf{k}} \frac{\epsilon_{\mathbf{q}\lambda}^\mu}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} (b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger) (e^{iq_\mu} - 1) (e^{ik_\mu} + e^{-ik'_\mu}) c_{\gamma, \mathbf{k}'}^\dagger c_{\gamma, \mathbf{k}} \sum_j \frac{1}{N} e^{i(\mathbf{q} + \mathbf{k} - \mathbf{k}') \cdot \mathbf{R}_j} = \quad (92)$$

$$= \frac{t'_\gamma}{\sqrt{N}} \sum_{\mathbf{k}, \mathbf{q}} \frac{\epsilon_{\mathbf{q}\lambda}^\mu}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} (b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger) (e^{iq_\mu} - 1) (e^{ik_\mu} + e^{-i(k_\mu + q_\mu)}) c_{\gamma, \mathbf{k} + \mathbf{q}}^\dagger c_{\gamma, \mathbf{k}} = \quad (93)$$

$$= \frac{2it'_\gamma}{\sqrt{N}} \sum_{\mathbf{k}, \mathbf{q}} \frac{\epsilon_{\mathbf{q}\lambda}^\mu}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} (\sin(k_\mu + q_\mu) - \sin(k_\mu)) c_{\gamma, \mathbf{k} + \mathbf{q}}^\dagger c_{\gamma, \mathbf{k}} (b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger) \quad (94)$$

Hamiltonka je na koncu oblike (intraband = $\gamma = \gamma'$):

$$H_{ep} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}, \mathbf{q}, \lambda} \psi_{\mathbf{k} + \mathbf{q}}^\dagger \begin{pmatrix} g_{aa}^\lambda(\mathbf{k}, \mathbf{q}) & g_{ab}^\lambda(\mathbf{q}) \\ g_{ba}^\lambda(\mathbf{q}) & g_{bb}^\lambda(\mathbf{k}, \mathbf{q}) \end{pmatrix} \psi_{\mathbf{k}} (b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger) \quad (95)$$

$$g^\lambda(\mathbf{k}, \mathbf{q}) = \frac{i}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \begin{pmatrix} 2t'_a T^\lambda(\mathbf{k}, \mathbf{q}) + F_{aa}^\lambda(\mathbf{q}) & F_{ab}^\lambda(\mathbf{q}) \\ F_{ba}^\lambda(\mathbf{q}) & 2t'_b T^\lambda(\mathbf{k}, \mathbf{q}) + F_{bb}^\lambda(\mathbf{q}) \end{pmatrix}. \quad (96)$$

kjer so:

$$T^\lambda(\mathbf{k}, \mathbf{q}) = \sum_{\mu=x, y} \epsilon_{\mathbf{q}\lambda}^\mu [\sin(k_\mu + q_\mu) - \sin(k_\mu)] \quad (97)$$

$$F_{\gamma\gamma'}^\lambda(\mathbf{q}) = D_{\gamma\gamma'}^{xx} \epsilon_{\mathbf{q}\lambda}^x \sin(q_x) + D_{\gamma\gamma'}^{yy} \epsilon_{\mathbf{q}\lambda}^y \sin(q_y) + D_{\gamma\gamma'}^{xy} [\epsilon_{\mathbf{q}\lambda}^y \sin(q_x) + \epsilon_{\mathbf{q}\lambda}^x \sin(q_y)]. \quad (98)$$

interband členi so neničelni, glede na izbrane orbitale (glej (87)) in izbran CF potencial, odvisno od simetrije. Ker smo izbrali $\mathbf{u}_{a,i} = \mathbf{u}_{b,i}$ se ti elementi sklapljajo samo z akustičnimi fononi, ce znotraj o.c. ni prostostnih stopenj(??).

2.4 Optični fononi

Rabimo dodatne ligande / dodatne ione znotraj celice. Še vedno obdržimo $\mathbf{u}_{a,j} = \mathbf{u}_{b,j} = \mathbf{u}_j$, okolica pa se deformira neodvisno. Odmik liganda ℓ v celici j razvijemo po normalnih nacinih

$$\mathbf{v}_{j,\ell} = \sum_s Q_{s,j} \mathbf{e}_{s,\ell}, \quad \sum_\ell \mathbf{e}_{s,\ell} \cdot \mathbf{e}_{s',\ell} = \delta_{ss'}, \quad (99)$$

kjer je s indeks nacina, $Q_{s,j}$ skalarna amplituda. Razvijemo enoelektronsko hamiltonko

$$h_j = h_0 + \sum_s \left. \frac{\partial U_{\text{el-ion}}}{\partial Q_{s,j}} \right|_{Q=0} Q_{s,j} + O(Q^2), \quad \Lambda_{\gamma\gamma'}^s = \left\langle \phi_\gamma \left| \frac{\partial U_{\text{el-ion}}}{\partial Q_{s,j}} \right| \phi_{\gamma'} \right\rangle_0. \quad (100)$$

$$H_{ep}^{\text{opt}} = \sum_{j,s} \sum_{\gamma, \gamma'} \Lambda_{\gamma\gamma'}^s Q_{s,j} c_{\gamma,j}^\dagger c_{\gamma',j}. \quad (101)$$

Kvantiziramo koordinate okoliških ionov:

$$Q_{s,j} = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}} \frac{1}{\sqrt{2M_s \omega_{\mathbf{q}s}}} e^{i\mathbf{q} \cdot \mathbf{R}_j} (b_{\mathbf{q}s} + b_{-\mathbf{q}s}^\dagger), \quad (102)$$

od tod

$$H_{ep}^{\text{opt}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}, \mathbf{q}, s} \psi_{\mathbf{k}+\mathbf{q}}^\dagger g^s(\mathbf{q}) \psi_{\mathbf{k}} \left(b_{\mathbf{q}s} + b_{-\mathbf{q}s}^\dagger \right), \quad g^s(\mathbf{q}) = \frac{\Lambda^s}{\sqrt{2M_s \omega_{\mathbf{q}s}}}. \quad (103)$$

Kar da končne sklopitve pri $\mathbf{q} \rightarrow 0$ ($\Omega_s \equiv \omega_{\mathbf{q} \rightarrow 0, s}$):

$$g^s(\mathbf{q} \rightarrow 0) = \frac{\Lambda^s}{\sqrt{2M_s \Omega_s}}. \quad (104)$$

2.5 Fotonske sklopitve v resonatorju

Začnemo z minimalnim modelom za elektrone:

$$H_e = H_e^{(0)} + H_V = \sum_k \varepsilon_{a,k} c_{a,k}^\dagger c_{a,k} + \sum_k \varepsilon_{b,k} c_{b,k}^\dagger c_{b,k} + V \sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \quad (105)$$

in privzamemo 2D plast, kjer sta dve orbitali na o.c., ki hibridizirata samo preko Coulombske interakcije. Disperzije za model neinteragirajoč sistem so v tem primeru enake (21). Interakcija pa je v prvem redu oblike:

$$H_{e\gamma}^{(1)} = - \int d\mathbf{r} \mathbf{j}_e \cdot \mathbf{A} = - \int d\mathbf{r} \sum_{\mathbf{q}} \mathbf{A}(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{r}} \mathbf{j}_e(\mathbf{r}) = - \sum_{\mathbf{q}} \mathbf{A}(\mathbf{q}) \cdot \mathbf{j}_e(-\mathbf{q}), \quad (106)$$

kjer je

$$\mathbf{A}(\mathbf{q}) = \sum_{\lambda} \frac{1}{\sqrt{2V\epsilon_0\omega_{\mathbf{q},\lambda}}} \epsilon_{\mathbf{q},\lambda} \left(a_{\mathbf{q},\lambda} + a_{-\mathbf{q},\lambda}^\dagger \right) \quad (107)$$

V t.b. modelu dobimo tok iz kontinuitetne in Heisenbergove enačbe:

$$\frac{\partial \rho(\mathbf{r})}{\partial t} + \nabla \cdot \mathbf{j} = 0 \rightarrow i\mathbf{q} \cdot \mathbf{j}(\mathbf{q}) = - \frac{\partial \rho(\mathbf{q})}{\partial t} \quad (108)$$

(+ je od IBP) in

$$\mathbf{q} \cdot \mathbf{j}_e(\mathbf{q}) = i \frac{\partial \rho(\mathbf{q})}{\partial t} = -[H_e, \rho(\mathbf{q})] \quad (109)$$

Elektronska gostota je

$$\rho(\mathbf{r}) = -e \cdot \psi^\dagger(\mathbf{r}) \psi(\mathbf{r}) \quad (110)$$

operatorje polja razvijemo po lokaliziranih / Wannierovih orbitalah

$$\psi(\mathbf{r}) = \sum_{i, \gamma=\{a,b\}} \phi_{\gamma}(\mathbf{r} - \mathbf{R}_i) c_{\gamma i}. \quad (111)$$

Transformiramo s FT, uvedemo $\mathbf{r} = \mathbf{R}_i + \mathbf{x}$ in razvijemo do dipolov $e^{-i\mathbf{q} \cdot \mathbf{x}} \simeq 1 - i\mathbf{q} \cdot \mathbf{x}$ (delamo se da se orbitale ne prekrivajo izven o.c.):

$$\rho(\mathbf{q}) = -e \sum_{i,j} \sum_{\gamma, \gamma'} \int d\mathbf{r} e^{-i\mathbf{q} \cdot \mathbf{r}} \phi_{\gamma}^*(\mathbf{r} - \mathbf{R}_i) \phi_{\gamma'}(\mathbf{r} - \mathbf{R}_j) c_{\gamma i}^\dagger c_{\gamma' j} = -e \sum_{i, \gamma, \gamma'} e^{-i\mathbf{q} \cdot \mathbf{R}_i} \langle \phi_{\gamma} | e^{-i\mathbf{q} \cdot \mathbf{x}} | \phi_{\gamma'} \rangle c_{\gamma i}^\dagger c_{\gamma' i} \quad (112)$$

Zdaj pa recimo da je dipolni moment samo med različnima orbitalama v o.c. in $n_{\gamma i} = \langle \phi_{\gamma} | \phi_{\gamma} \rangle c_{\gamma i}^\dagger c_{\gamma i}$:

$$\rho(\mathbf{q}) \simeq -e \sum_i e^{-i\mathbf{q} \cdot \mathbf{R}_i} \left(n_{ai} + n_{bi} - i\mathbf{q} \cdot \mathbf{d} \left(c_{a,i}^\dagger c_{b,i} + c_{b,i}^\dagger c_{a,i} \right) \right) \quad (113)$$

zdaj uporabimo zvezo

$$\sum_i e^{i\mathbf{q} \cdot \mathbf{R}_i} c_{ai}^\dagger c_{bi} = \sum_{\mathbf{k}} c_{a, \mathbf{k}+\mathbf{q}}^\dagger c_{b\mathbf{k}} \quad (114)$$

z negativnim argumentom je potem elektronska gostota do dipolnega člena (upoštevajoč $\mathbf{d} = \langle \phi_a(\mathbf{x}) | \mathbf{x} | \phi_b(\mathbf{x}) \rangle$):

$$\rho(-\mathbf{q}) = -e \sum_{\mathbf{k}} \begin{pmatrix} c_{a, \mathbf{k}+\mathbf{q}}^\dagger & c_{b, \mathbf{k}+\mathbf{q}}^\dagger \end{pmatrix} \left(\mathbb{1} + i\mathbf{q} \cdot \mathbf{d} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right) \begin{pmatrix} c_{a\mathbf{k}} \\ c_{b\mathbf{k}} \end{pmatrix} \quad (115)$$

Zdaj poračunamo komutator, da dobimo tok. **Lahko vzamemo H_e^0 , ker za local V se členi z $\Delta_k \rightarrow \Delta$ pokrajsajo!!**, načeloma pa (nisem še preveru, z dipolnim členom ne komutira). Delam z vektorji $\psi_k = (c_{a,k} \ c_{b,k})^\top$:

$$[H_e^{(0)}, \rho(\mathbf{q})] = -e \sum_{\mathbf{k}, \mathbf{k}'} \left[\psi_{\mathbf{k}}^\dagger \begin{pmatrix} \varepsilon_{a,\mathbf{k}} & 0 \\ 0 & \varepsilon_{b,\mathbf{k}} \end{pmatrix} \psi_{\mathbf{k}}, \psi_{\mathbf{k}'-\mathbf{q}}^\dagger \begin{pmatrix} 1 & -i\mathbf{q} \cdot \mathbf{d} \\ -i\mathbf{q} \cdot \mathbf{d} & 1 \end{pmatrix} \psi_{\mathbf{k}'} \right] \quad (116)$$

uporabimo zvezo:

$$\left[\psi_{\mathbf{k}}^\dagger A \psi_{\mathbf{k}}, \psi_{\mathbf{k}'-\mathbf{q}}^\dagger B \psi_{\mathbf{k}'} \right] = \psi_{\mathbf{k}'-\mathbf{q}}^\dagger \left(A(\mathbf{k}' - \mathbf{q}) B(\mathbf{k}') - B(\mathbf{k}' - \mathbf{q}) A(\mathbf{k}') \right) \psi_{\mathbf{k}'} \delta_{\mathbf{k}, \mathbf{k}'-\mathbf{q}} \quad (117)$$

$$= \psi_{\mathbf{k}}^\dagger \left(A(\mathbf{k}) B(\mathbf{k} + \mathbf{q}) - B(\mathbf{k}) A(\mathbf{k} + \mathbf{q}) \right) \psi_{\mathbf{k}+\mathbf{q}} \quad (118)$$

od tod

$$\mathbf{q} \cdot \mathbf{j}_e(\mathbf{q}) = -[H_e^{(0)}, \rho(\mathbf{q})] = -e \sum_{\mathbf{k}} \psi_{\mathbf{k}}^\dagger \begin{pmatrix} \varepsilon_{a,\mathbf{k}+\mathbf{q}} - \varepsilon_{a,\mathbf{k}} & i(\mathbf{q} \cdot \mathbf{d})(\varepsilon_{a,\mathbf{k}} - \varepsilon_{b,\mathbf{k}+\mathbf{q}}) \\ i(\mathbf{q} \cdot \mathbf{d})(\varepsilon_{b,\mathbf{k}} - \varepsilon_{a,\mathbf{k}+\mathbf{q}}) & \varepsilon_{b,\mathbf{k}+\mathbf{q}} - \varepsilon_{b,\mathbf{k}} \end{pmatrix} \psi_{\mathbf{k}+\mathbf{q}} \quad (119)$$

da izrazimo tok, naredimo približek, da prispevajo le majhni $\mathbf{q} \rightarrow 0$, zato:

$$\varepsilon_{a,\mathbf{k}+\mathbf{q}} - \varepsilon_{a,\mathbf{k}} \simeq \mathbf{q} \cdot \nabla_{\mathbf{k}} \varepsilon_{a,\mathbf{k}+\mathbf{q}/2}, \quad (120)$$

kar drži točno za proste elektrone in približno za tight binding disperzije. In general ce ne expandamo:

$$\mathbf{j}_e(\mathbf{q}) = -\frac{e\mathbf{q}}{\|\mathbf{q}\|^2} \sum_{\mathbf{k}} \psi_{\mathbf{k}}^\dagger M(\mathbf{k}, \mathbf{q}) \psi_{\mathbf{k}+\mathbf{q}} + \mathbf{q} \times \mathbf{C}(\mathbf{q}) \quad (121)$$

kjer so $\mathbf{C}(\mathbf{q})$ arbitrarna vektorska polja (tokovi brez divergence, magnetizacijski tokovi). Ce pa naredimo ta približek pa dobimo za tok v splošnem, indekse zamaknemo za $-\mathbf{q}$:

$$\mathbf{j}_e(\mathbf{q}) \simeq -e \sum_{\mathbf{k}} \psi_{\mathbf{k}-\mathbf{q}}^\dagger \begin{pmatrix} \nabla_{\mathbf{k}} \varepsilon_{a,\mathbf{k}-\mathbf{q}/2} & i\mathbf{d}(\varepsilon_{a,\mathbf{k}-\mathbf{q}} - \varepsilon_{b,\mathbf{k}}) \\ i\mathbf{d}(\varepsilon_{b,\mathbf{k}-\mathbf{q}} - \varepsilon_{a,\mathbf{k}}) & \nabla_{\mathbf{k}} \varepsilon_{b,\mathbf{k}-\mathbf{q}/2} \end{pmatrix} \psi_{\mathbf{k}}. \quad (122)$$

Od tod splošna Hamiltonka v prvem redu:

$$H_{e\gamma}^{(1)} = - \sum_{\mathbf{q}} \mathbf{j}_e(-\mathbf{q}) \cdot \mathbf{A}(\mathbf{q}) = \quad (123)$$

$$= \sum_{\mathbf{k}, \mathbf{q}, \lambda} \frac{e}{\sqrt{2\mathcal{V}\varepsilon_0\omega_{\mathbf{q},\lambda}}} \psi_{\mathbf{k}+\mathbf{q}}^\dagger \underbrace{\begin{pmatrix} \epsilon_{\mathbf{q}\lambda} \cdot \mathbf{v}_a(\mathbf{k} + \frac{\mathbf{q}}{2}) & i(\epsilon_{\mathbf{q}\lambda} \cdot \mathbf{d})(\varepsilon_{a,\mathbf{k}+\mathbf{q}} - \varepsilon_{b,\mathbf{k}}) \\ i(\epsilon_{\mathbf{q}\lambda} \cdot \mathbf{d})(\varepsilon_{b,\mathbf{k}+\mathbf{q}} - \varepsilon_{a,\mathbf{k}}) & \epsilon_{\mathbf{q}\lambda} \cdot \mathbf{v}_b(\mathbf{k} + \frac{\mathbf{q}}{2}) \end{pmatrix}}_{M^\lambda(\mathbf{k}, \mathbf{q})} \psi_{\mathbf{k}} \left(a_{\mathbf{q}\lambda} + a_{-\mathbf{q}\lambda}^\dagger \right). \quad (124)$$

Za setup od Curtisa: material 2D v xy ravnini, stoječe EM valovanje v z , vzamemo osnovni mode ($n = 1$), ki prispeva največ, v tem primeru sta dve možni polarizaciji:

$$\epsilon_{\text{TE}}(z) = (\hat{\mathbf{z}} \times \hat{\mathbf{q}}) \sin\left(\frac{\pi}{L}z\right) \quad (125)$$

$$\epsilon_{\text{TM}}(z) = \frac{\omega_0}{\omega_{\mathbf{q}}} \hat{\mathbf{q}} \sin\left(\frac{\pi}{L}z\right) + i \frac{c}{\omega_{\mathbf{q}}} \hat{\mathbf{z}} \cos\left(\frac{\pi}{L}z\right) \quad (126)$$

kjer je $\mathbf{q} = q(\cos \varphi, \sin \varphi)$ valovni vektor v ravnini, $\omega_0 = \pi c/L$ in $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$. Na sredini resonatorja, pri $z = L/2$, je $\sin(\pi z/L) = 1$ in $\cos(\pi z/L) = 0$, zato:

$$\epsilon_{\text{TE}}\left(\frac{L}{2}\right) = \hat{\mathbf{z}} \times \hat{\mathbf{q}} = \begin{pmatrix} -\sin \varphi \\ \cos \varphi \\ 0 \end{pmatrix} \quad (127)$$

Nacin	$\epsilon \cdot \mathbf{v}_\gamma \left(\mathbf{k} + \frac{\mathbf{q}}{2} \right)$	$\epsilon \cdot \mathbf{d}, \mathbf{d}$ in plane	$\epsilon \cdot \mathbf{d}, \mathbf{d} \parallel \hat{\mathbf{z}}$
TE	$2t_\gamma \left(-\sin \varphi \sin \left(k_x + \frac{q_x}{2} \right) + \cos \varphi \sin \left(k_y + \frac{q_y}{2} \right) \right)$	$d \sin(\theta - \varphi)$	0
TM	$\frac{\omega_0}{\omega_q} 2t_\gamma \left(\cos \varphi \sin \left(k_x + \frac{q_x}{2} \right) + \sin \varphi \sin \left(k_y + \frac{q_y}{2} \right) \right)$	$\frac{\omega_0}{\omega_q} d \cos(\theta - \varphi)$	$i d \frac{c q}{\omega_q} \cos \left(\frac{\pi}{L} z \right) (= 0 \text{ pri } z = \frac{L}{2})$

$$\epsilon_{\text{TM}} \left(\frac{L}{2} \right) = \frac{\omega_0}{\omega_q} \hat{\mathbf{q}} = \frac{\omega_0}{\omega_q} \begin{pmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{pmatrix} \quad (128)$$

Iz tega imamo pri $z = L/2$ možne sklopitve glede na model ($\mathbf{d} = d(\cos \theta, \sin \theta)$ v ravnini oz. $\mathbf{d} = d \hat{\mathbf{z}}$ izven ravnine): dipolni moment je neničeln za liho-sode kombinacije orbital in plane, tega gledamo samo glede na o.c.:

$$\mathbf{d} = \langle \phi_a | \mathbf{x} | \phi_b \rangle = \int \phi_a^*(\mathbf{x}) \mathbf{x} \phi_b(\mathbf{x}) d\mathbf{x} \quad (129)$$

2.5.1 Model prostih elektronov in sklopitev s fotoni

Kaj pa če gledamo namesto tight binding, proste elektrone, torej

$$\varepsilon_{\gamma \mathbf{k}} = \pm \frac{G}{2} \mp 2t(\cos k_x + \cos k_y) \quad \text{za } k_x, k_y \sim 0 \quad (130)$$

od tod

$$\varepsilon_{\gamma \mathbf{k}} \approx \pm \frac{G}{2} \mp 2t \left(1 - \frac{k_x^2}{2} + 1 - \frac{k_y^2}{2} \right) = \pm \left(\frac{G}{2} - 4t \right) \pm t(k_x^2 + k_y^2) = \pm(\varepsilon^0 + t\mathbf{k} \cdot \mathbf{k}) \quad (131)$$

Torej je Hamiltonka

$$H_e^{(0)} = \sum_{\gamma, \mathbf{k}} (\varepsilon_\gamma^0 + t_\gamma \mathbf{k}^2) c_{\gamma, \mathbf{k}}^\dagger c_{\gamma, \mathbf{k}} \quad (132)$$

Na drug način lahko poračunamo tok, ker razvijemo

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{k}, \gamma=\{a,b\}} e^{i\mathbf{k} \cdot \mathbf{r}} c_{\gamma, \mathbf{k}}. \quad (133)$$

in za tok uporabimo

$$\mathbf{j}_e(\mathbf{r}) = -\frac{e}{2mi} (\psi^\dagger(\mathbf{r}) \nabla \psi(\mathbf{r}) - \nabla \psi^\dagger(\mathbf{r}) \psi(\mathbf{r})) = -\frac{e}{2mi} \frac{1}{\mathcal{V}} \sum_{\gamma, \mathbf{k}, \mathbf{k}'} \left(c_{\gamma \mathbf{k}'}^\dagger c_{\gamma \mathbf{k}} e^{-i\mathbf{k}' \cdot \mathbf{r}} (i\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}} - c_{\gamma \mathbf{k}'}^\dagger c_{\gamma \mathbf{k}} (-i\mathbf{k}') e^{-i\mathbf{k}' \cdot \mathbf{r}} e^{i\mathbf{k} \cdot \mathbf{r}} \right) \quad (134)$$

s FT dobimo

$$\mathbf{j}_e(-\mathbf{q}) = \int d\mathbf{r} \mathbf{j}_e(\mathbf{r}) e^{i\mathbf{q} \cdot \mathbf{r}} = -\frac{e}{2m} \sum_{\gamma, \mathbf{k}} \left(c_{\gamma, \mathbf{k}+\mathbf{q}}^\dagger c_{\gamma \mathbf{k}} \mathbf{k} + (\mathbf{k} + \mathbf{q}) c_{\gamma, \mathbf{k}+\mathbf{q}}^\dagger c_{\gamma \mathbf{k}} \right) = -\frac{e}{m} \sum_{\mathbf{k}} \left(\mathbf{k} + \frac{\mathbf{q}}{2} \right) \psi_{\mathbf{k}+\mathbf{q}}^\dagger \psi_{\mathbf{k}} \quad (135)$$

kar bi dobili tudi iz bolj splošne relacije. V tem primeru dobimo enako kot Curtis, $m = 1/2t$

$$H_{e\gamma}^{(1)} = \frac{e}{m} \sum_{\mathbf{k}, \mathbf{q}, \lambda} \frac{1}{\sqrt{2\mathcal{V}\epsilon_0\omega_{\mathbf{q},\lambda}}} \left(\mathbf{k} + \frac{\mathbf{q}}{2} \right) \cdot \epsilon_{\mathbf{q}\lambda} \psi_{\mathbf{k}+\mathbf{q}}^\dagger \psi_{\mathbf{k}} (a_{\mathbf{q}\lambda} + a_{-\mathbf{q}\lambda}^\dagger) \quad (136)$$

2.5.2 Drugi red sklopitve za t.b.

Zacnemo spet:

$$H_e^{(0)} = \sum_{\gamma, j} \varepsilon_\gamma^{(0)} c_{\gamma j}^\dagger c_{\gamma j} - \sum_{j, \delta, \gamma} t_\gamma \left(c_{\gamma j}^\dagger c_{\gamma j+\delta} + \text{h.c.} \right) \quad (137)$$

v prisotnosti polja dobi prekrivalni integral fazo:

$$t_\gamma \rightarrow t_\gamma e^{i\theta_{j\delta}} \quad \theta_{j\delta} = e \int_{\mathbf{R}_j}^{\mathbf{R}_j+\delta} \mathbf{A}(\mathbf{r}, t) \cdot d\mathbf{r} \quad (138)$$

od tod

$$H_e^{(0)}[\mathbf{A}] = \sum_{\gamma,j} \varepsilon_{\gamma}^{(0)} c_{\gamma j}^{\dagger} c_{\gamma j} - \sum_{j,\delta,\gamma} t_{\gamma} \left(e^{i\theta_j \delta} c_{\gamma j}^{\dagger} c_{\gamma j+\delta} + e^{-i\theta_j \delta} c_{\gamma j+\delta}^{\dagger} c_{\gamma j} \right) \quad (139)$$

od kjer razvijemo do drugega reda, prvi red bi moral biti isti kot gor (razen dipolov)

$$e^{i\theta} = 1 + \underbrace{i\theta}_{\mathcal{O}(eA)} - \underbrace{\frac{1}{2}\theta^2}_{\mathcal{O}(e^2 A^2)} \quad (140)$$

vstavimo

$$c_{\gamma j} = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{R}_j} c_{\gamma \mathbf{k}} \quad (141)$$

in približek da se polje ne spremeni veliko do najbližjih sosedov

$$\int_{\mathbf{R}_j}^{\mathbf{R}_j+\delta} \mathbf{A}(\mathbf{r}, t) \cdot d\mathbf{r} \approx \mathbf{A} \left(\mathbf{R}_j + \frac{\delta}{2} \right) \cdot \delta \quad (\mathbf{A} \text{ se ne spremeni veliko znotraj celice+midpoint approx.}) \quad (142)$$

računamo

$$H_{e\gamma}^{(2)} = - \sum_{j,\delta,\gamma} t_{\gamma} \left(-\frac{\theta_j^2 \delta}{2} \right) \left(c_{\gamma j}^{\dagger} c_{\gamma j+\delta} + c_{\gamma j+\delta}^{\dagger} c_{\gamma j} \right) = \quad (143)$$

$$= \sum_{j,\delta,\gamma} t_{\gamma} \cdot \frac{e^2}{2} \int_{\mathbf{R}_j}^{\mathbf{R}_j+\delta} \mathbf{A}(\mathbf{r}, t) \cdot d\mathbf{r} \int_{\mathbf{R}_j}^{\mathbf{R}_j+\delta} \mathbf{A}(\mathbf{r}, t) \cdot d\mathbf{r} \left(c_{\gamma j}^{\dagger} c_{\gamma j+\delta} + \text{h.c.} \right) = \quad (144)$$

$$= \sum_{j,\delta,\gamma} t_{\gamma} \frac{e^2}{2} \left(\sum_{\mathbf{q},\lambda} \frac{1}{\sqrt{2\varepsilon_0 \mathcal{V} \omega_{\mathbf{q}\lambda}}} (\varepsilon_{\mathbf{q}\lambda} \cdot \delta) e^{i\mathbf{q} \cdot (\mathbf{R}_j + \frac{\delta}{2})} (b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^{\dagger}) \right) \left(\sum_{\mathbf{q}',\lambda'} \frac{1}{\sqrt{2\varepsilon_0 \mathcal{V} \omega_{\mathbf{q}'\lambda'}}} (\varepsilon_{\mathbf{q}'\lambda'} \cdot \delta) e^{i\mathbf{q}' \cdot (\mathbf{R}_j + \frac{\delta}{2})} (b_{\mathbf{q}'\lambda'} + b_{-\mathbf{q}'\lambda'}^{\dagger}) \right) \quad (145)$$

$$\cdot \sum_{\mathbf{k},\mathbf{k}'} \frac{1}{\mathcal{V}} c_{\gamma \mathbf{k}'}^{\dagger} c_{\gamma \mathbf{k}} e^{-i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{R}_j} \left(e^{i\mathbf{k} \cdot \delta} + e^{i\mathbf{k}' \cdot \delta} \right) \quad (146)$$

ven pade ohranitev momenta za dvofotonske procese

$$\sum_j \frac{1}{\mathcal{V}} e^{i(\mathbf{q} + \mathbf{q}' - \mathbf{k}' + \mathbf{k}) \cdot \mathbf{R}_j} = \delta_{\mathbf{k}', \mathbf{k} + \mathbf{q} + \mathbf{q}'} \quad (147)$$

kar da na koncu

$$= \sum_{\substack{\gamma, \mathbf{k}, \mathbf{q}, \mathbf{q}' \\ \lambda, \lambda', \delta}} t_{\gamma} \frac{e^2}{2} \frac{1}{2\varepsilon_0 \mathcal{V} \sqrt{\omega_{\mathbf{q}\lambda} \omega_{\mathbf{q}'\lambda'}}} (\varepsilon_{\mathbf{q}\lambda} \cdot \delta) (\varepsilon_{\mathbf{q}'\lambda'} \cdot \delta) (b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^{\dagger}) (b_{\mathbf{q}'\lambda'} + b_{-\mathbf{q}'\lambda'}^{\dagger}) c_{\gamma, \mathbf{k} + \mathbf{q} + \mathbf{q}'}^{\dagger} c_{\gamma \mathbf{k}} \quad (148)$$

$$\cdot \left(e^{i\mathbf{k} \cdot \delta} + e^{-i(\mathbf{k} + \mathbf{q} + \mathbf{q}') \cdot \delta} \right) \cdot e^{i(\mathbf{q} + \mathbf{q}') \cdot \frac{\delta}{2}} \quad (149)$$

POenostavimo zadnji člen v kosinus:

$$H_{e\gamma}^{(2)} = \sum_{\substack{\mathbf{k}, \mathbf{q}, \mathbf{q}' \\ \lambda, \lambda'}} \psi_{\mathbf{k} + \mathbf{q} + \mathbf{q}'}^{\dagger} M^{\lambda\lambda'}(\mathbf{k}, \mathbf{q}, \mathbf{q}') \psi_{\mathbf{k}} \cdot (a_{\mathbf{q},\lambda} + a_{-\mathbf{q},\lambda}^{\dagger})(a_{\mathbf{q}',\lambda'} + a_{-\mathbf{q}',\lambda'}^{\dagger}) \quad (150)$$

$$M^{\lambda\lambda'}(\mathbf{k}, \mathbf{q}, \mathbf{q}') = \sum_{\delta} \frac{e^2}{2\varepsilon_0 \mathcal{V} \sqrt{\omega_{\mathbf{q},\lambda} \omega_{\mathbf{q}',\lambda'}}} (\epsilon_{\mathbf{q},\lambda} \cdot \delta) (\epsilon_{\mathbf{q}',\lambda'} \cdot \delta) \cos \left(\left(\mathbf{k} + \frac{\mathbf{q} + \mathbf{q}'}{2} \right) \cdot \delta \right) \begin{pmatrix} t_a & 0 \\ 0 & t_b \end{pmatrix} \quad (151)$$

Za n.n. v square latticu eksplisitno:

$$M^{\lambda\lambda'}(\mathbf{k}, \mathbf{q}, \mathbf{q}') = \frac{e^2}{2\varepsilon_0 \mathcal{V} \sqrt{\omega_{\mathbf{q},\lambda} \omega_{\mathbf{q}',\lambda'}}} \left[\epsilon_{\mathbf{q},\lambda}^x \epsilon_{\mathbf{q}',\lambda'}^x \cos\left(k_x + \frac{q_x + q'_x}{2}\right) + \epsilon_{\mathbf{q},\lambda}^y \epsilon_{\mathbf{q}',\lambda'}^y \cos\left(k_y + \frac{q_y + q'_y}{2}\right) \right] \begin{pmatrix} t_a & 0 \\ 0 & t_b \end{pmatrix}. \quad (152)$$

A je to consistent z $\mathcal{O}(eA)$? Če ja, kako dobimo dipole notri v te elemente, al se sam ne sklapljajo nikoli?

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